

Bangladesh University of Engineering and Technology

Department of Chemistry

CHEM113: Chemistry

For Students of CSE

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CHEM113 : Chemistry

Total Credit Hours: 3 ; L1/T2

Course Content

Quantum concept in atomic structure, VSEPR; molecular geometry, Quantum concept in bonding; VBT and MOT, Frontier MOT and electronic transition, Silicon chemistry, Properties of solutions, Colloid and Nano-chemistry, ~~Phase rule and phase diagram; Energy and chemistry, Electrochemistry; electrolytic conduction, corrosion, devices for energy storage,~~ **Chemistry of biodegradable and conductive polymer; LED, LCD/touch screen,** Chemistry of proteins, nucleic acids (DNA, RNA), carbohydrates and lipids; Introduction to computational chemistry; Design of new molecules, materials and drug.

Polymers: Definition and importance

Fundamental of polymers

Macromolecules and polymers, Monomer, Type of polymers, Oligomers

Conducting polymers

Special properties, Importance, Conductivity, Classification, Preparation

Polymers

What are Polymers and Why Polymers are Important?

Long Chain Molecules

Extraordinary Range of Physical Properties

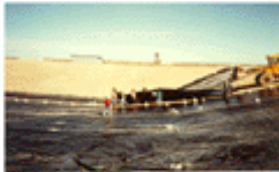
Many (Not All) are Cheap

We live in
a polymer
age

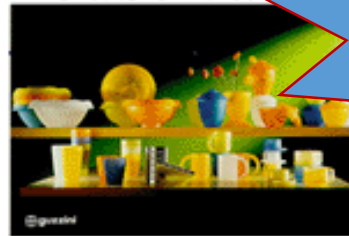
Synthetic Polymers



Geomembranes



Houseware



Fibres



Wire & Cable



PVC



POLYSTYRENE



PET



POLYURETHANE



PERSPEX

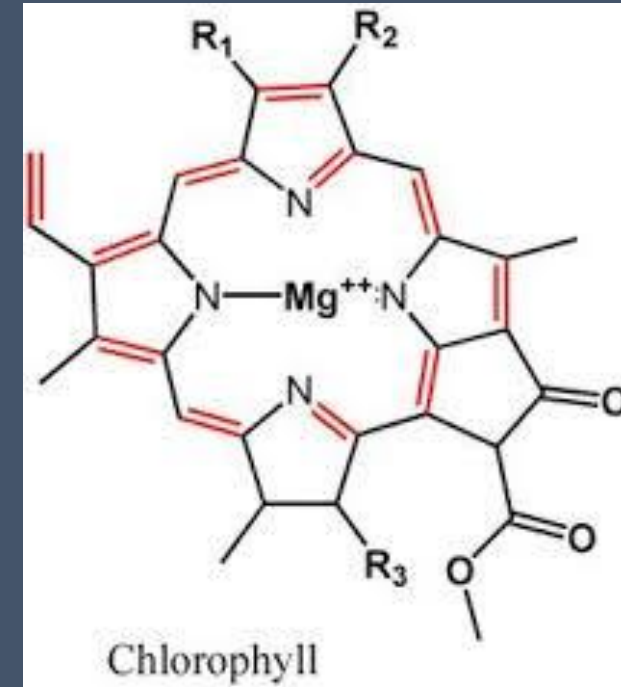


POLYCARBONATES

Macromolecules and Polymers

Macromolecules-

- also called polymers
- may *NOT* contain monomer unit
- all polymers are macromolecules, but all macromolecules are not polymers



Many repeating units

POLYMER

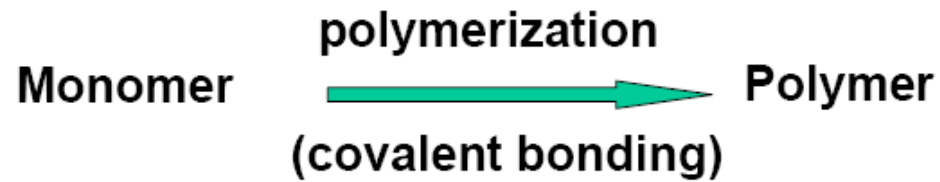
A large molecule made up of small building blocks (monomers)

MONOMERS

Building blocks

Polymers-

Polymerization, Oligomers, Polymer type



Greek mono + mer poly + mer
 single part many part

Monomer $\Rightarrow$ Oligomer $\Rightarrow$ Polymer

oligos + mer
few part

HOMOPOLYMER

What you get if the building blocks are all **the same**

COPOLYMER

A polymer made up of **different monomers**

BLEND

A mixture of **different polymers**

-A-A-A-A-A-A-A-

Homopolymer

-A-B-B-A-B-A-A-B-

Random copolymer

-A-B-A-B-A-B-A-B-

Alternating copolymer

-A-A-A-A-B-B-B-B-

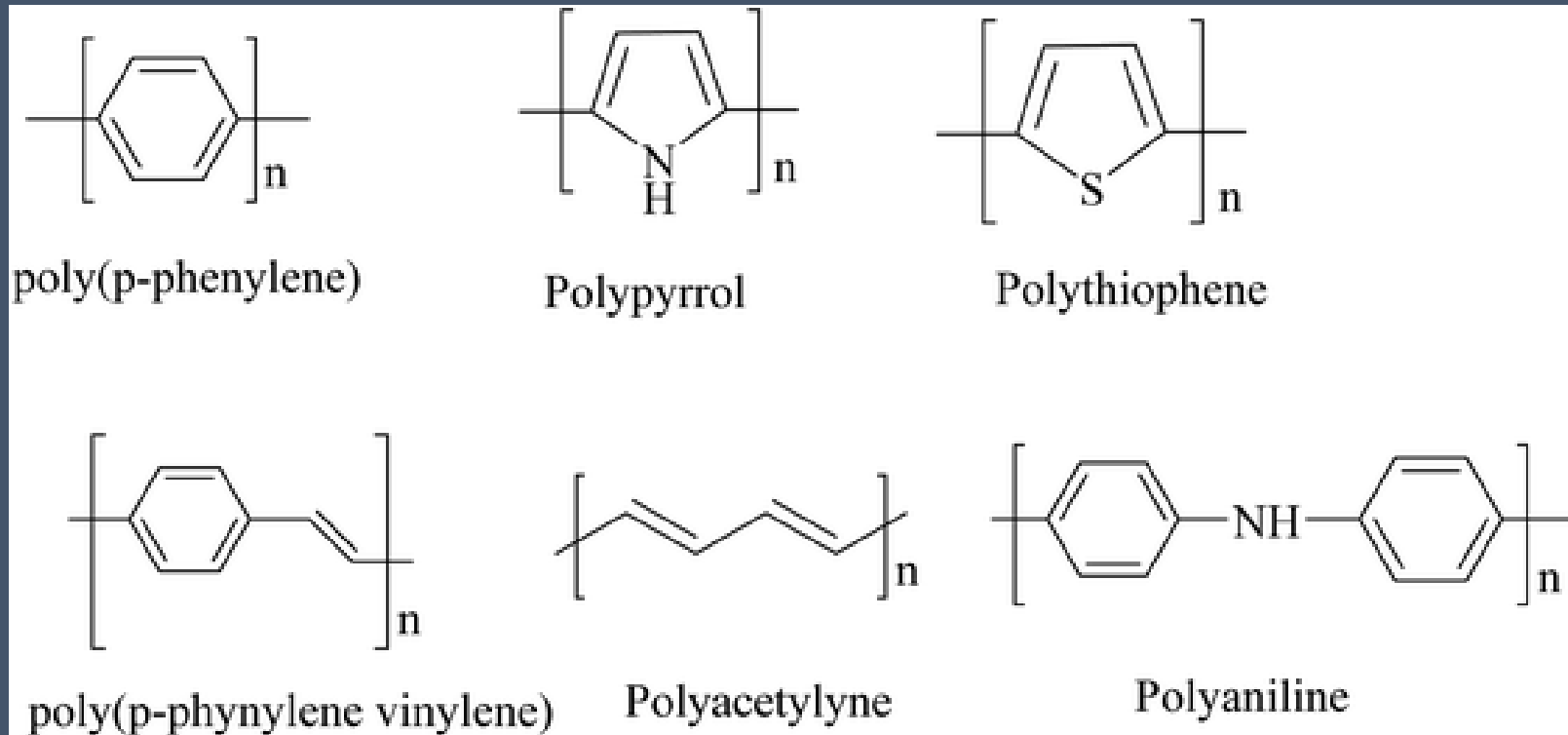
Block copolymer

-A-A-A-A-A-A-A-
 |
 B-B-B-B-B-B-

Graft copolymer

Conducting polymers vs. Conduction

- Conducting polymers are organic polymers that can conduct electricity, exhibiting properties similar to **semiconductors and even metals**.
- They are characterized by their ability to transport charge through their conjugated polymer chains.



Conducting polymers_ Importance

Conducting polymers are significant because they offer a unique combination of polymer properties (lightweight, flexibility, ease of processing) with the electrical conductivity of metals. This makes them valuable in a wide range of applications, including electronics, sensors, and energy storage.

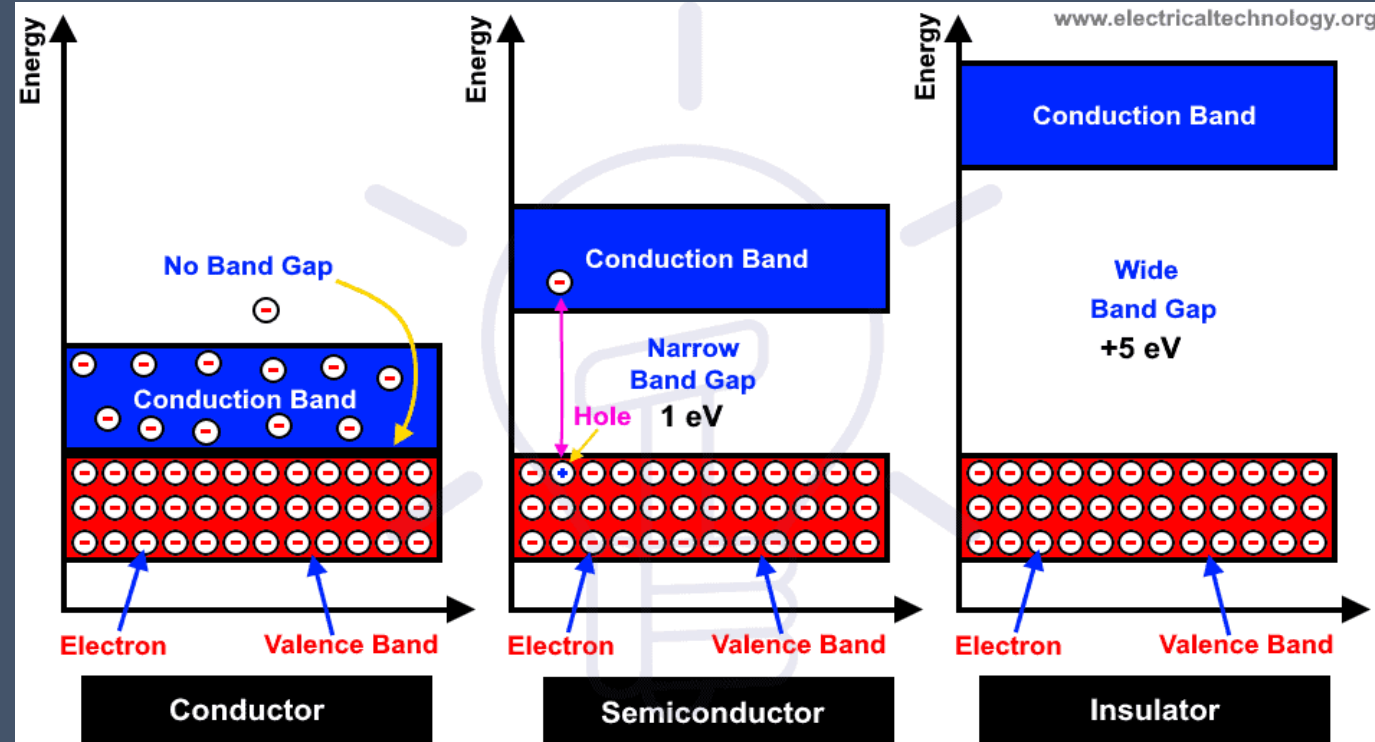
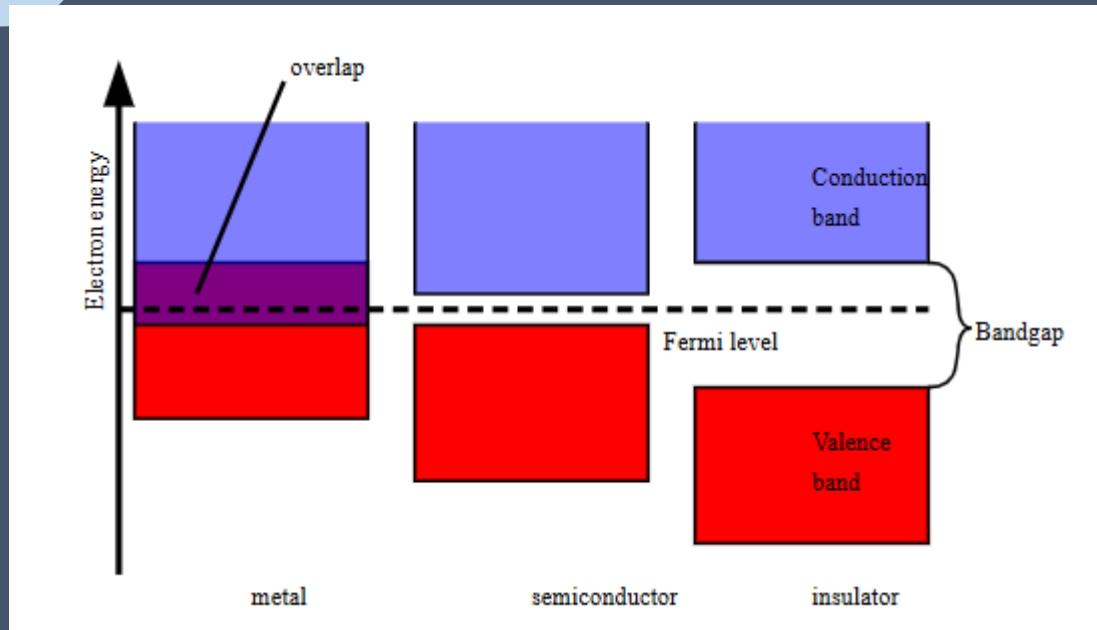
Unique properties of conducting polymers

- **Lightweight and Flexible:** Unlike traditional metals, conducting polymers are lightweight and flexible, allowing for diverse applications in wearable technology, flexible displays, and other areas where materials need to conform to shapes.
- **Tunable Conductivity:** Their conductivity can be adjusted through doping, enabling precise control over electrical properties for specific applications.
- **Potential for Chemical Sensing:** Conducting polymers can be designed to change their electrical properties in response to specific chemicals or environmental conditions, making them useful in sensors.
- **Biocompatibility:** Some conducting polymers exhibit biocompatibility, opening possibilities for use in biomedical devices and implants.

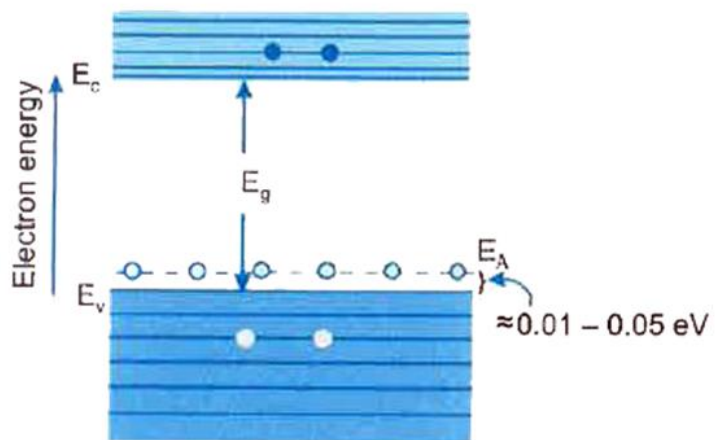
Conducting polymers_ Key Applications

- ❑ **Electronics:** Conducting polymers are used in transistors, LEDs, solar cells, and other electronic components.
- ❑ **Sensors:** They are employed in various sensors, such as gas sensors, biosensors, and chemical sensors.
- ❑ **Energy Storage:** They play a role in rechargeable batteries and supercapacitors.
- ❑ **Corrosion Protection:** Conducting polymers can be used as protective coatings for metals.
- ❑ **Displays:** They are utilized in displays, particularly in flexible and transparent applications.
- ❑ **Antistatic Coatings:** Conducting polymers can provide antistatic (preventing the build-up of static electricity) properties for various materials.
- ❑ **Biomedical Applications:** They are being explored for drug delivery, tissue engineering, and other biomedical uses.

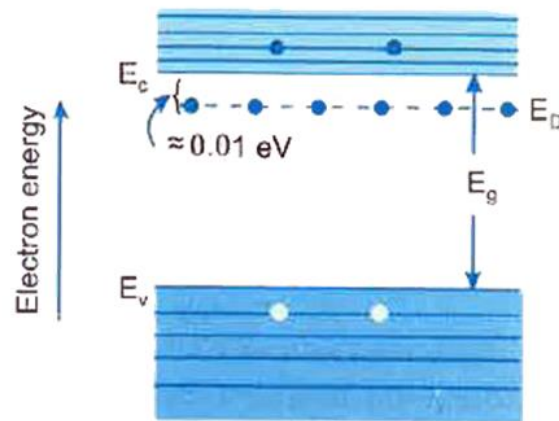
Conducting polymers vs. Conduction



p-type semiconductor



n-type semiconductor



Classification of conducting polymers: Composition

Depending on Composition

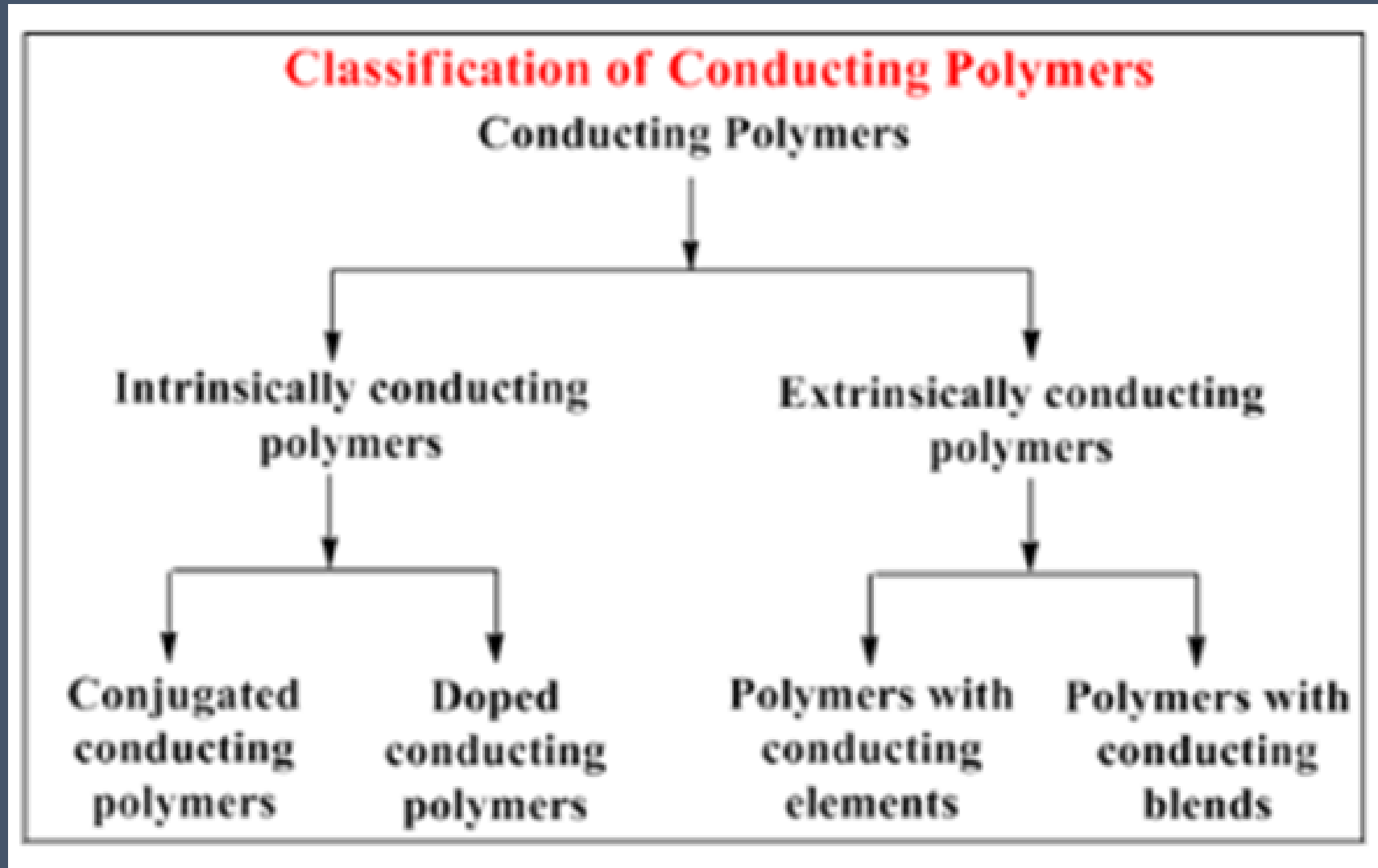
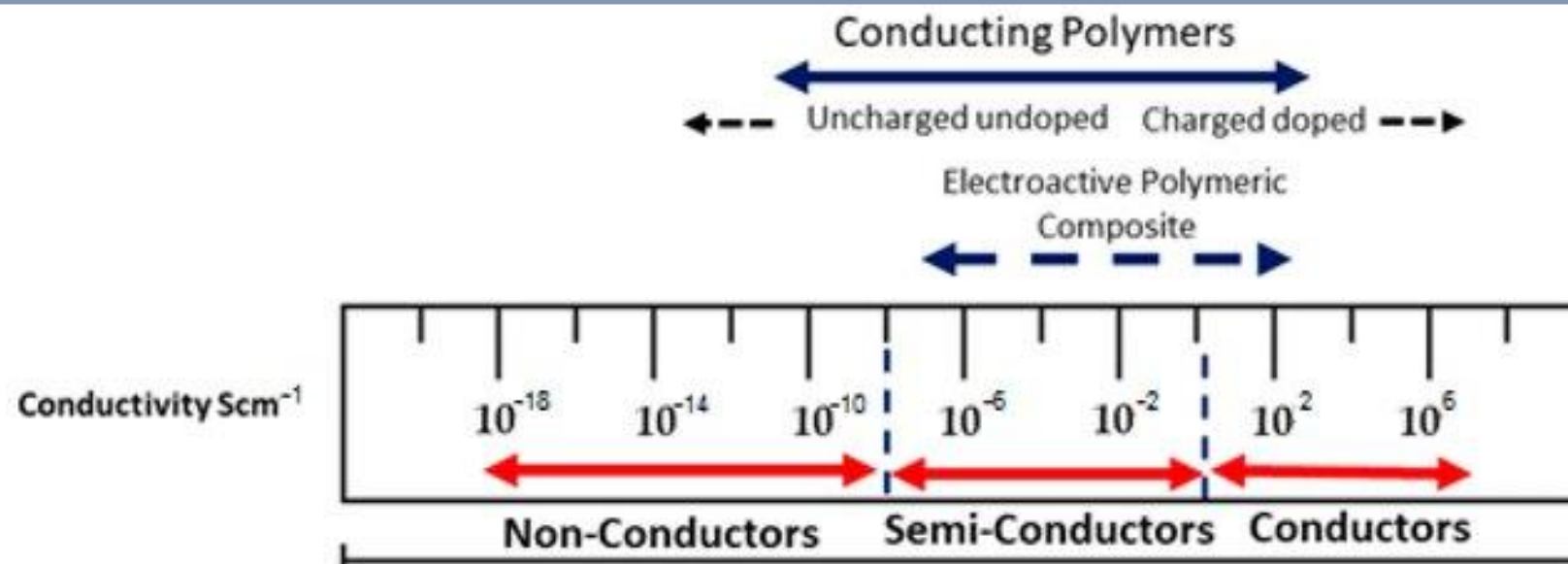


Table 4.5



<i>Material</i>	<i>Conductivity (S/cm)^b</i>
Copper	5.8×10^5
Gold	4.1×10^5
Polyacetylene	$10^3 - 10^5$
Poly(sulfur nitride)	$10^3 - 10^4$
Poly(<i>p</i> -phenylene)	10^3
Poly(<i>p</i> -phenylenevinylene)	10^3
Polyaniline	$10^2 - 10^3$
Polypyrrole	$10^2 - 10^3$

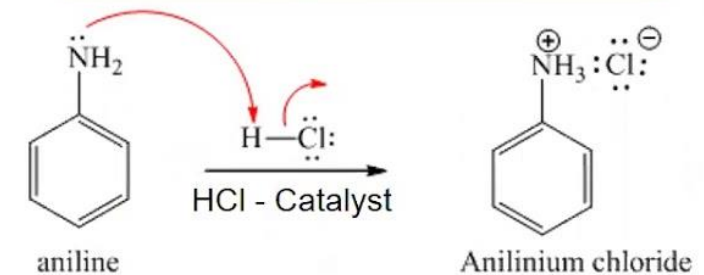
{ Insulators: $\sigma < 10^{-8} \text{ S/cm}$
 Semiconductors: $10^{-7} < \sigma < 10^{-1} \text{ S/cm}$
 Conductors: $\sigma > 10^2 \text{ S/cm}$

Synthesis of Polyaniline

Polyaniline Preparation



Polyaniline polymerization reaction



Polyaniline is synthesized by oxidative polymerization of aniline

Stirring at $\sim 0^\circ\text{C}$ $\downarrow$ $(\text{NH}_4)_2\text{S}_2\text{O}_8$ Oxidant



Factors influencing the conductivity of polymers

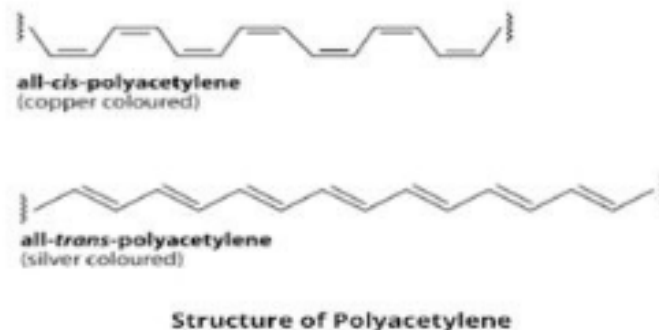
- ❑ **Chain length:** Increases with the increase in the length of the polymer chain
- ❑ **Doping level.** Increases with the increase in doping level till the saturation point
- ❑ **Charge carriers:** When charge carriers are introduced into the conduction band or valence band the electrical conductivity increase dramatically
- ❑ **Conjugation:** Conjugation (polymer backbone consists of alternating single and double bond. Increases with increasing number of conjugations

Intrinsically Conducting Polymers

- Intrinsically conducting polymers are substances which have a π -bond backbone.
- These π -electrons flow from one point to another in the polymer which results to their ability to conduct electricity.
- The conjugation can be due to either π electrons or due to doped ingredients.
- On the basis of above point they can be further classified into:
 1. Conjugated Conducting Polymers
 2. Doped Conducting Polymers

Conjugated Conducting Polymers

- In these types of polymers, due to the presence of double bonds and lone pair of electrons conduction of electricity takes place.
- Due to overlapping of conjugated π electrons, valence and conduction bands throughout the backbone of the polymer results into conduction of electricity.
- Electrical conduction can occur only after attainment of required energy of activation either thermally or photochemically because there is some gap between the valence and conduction bands. So the electrons need to be excited by some means.
- Ex: Polyacetylene, polyaniline, etc.

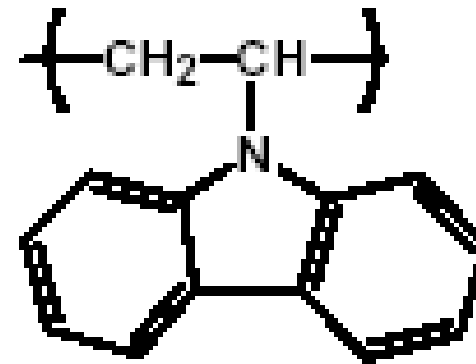


Doped Conducting Polymers

- These polymers are made by adding some foreign material or desired impurities which enhance their conduction power.
- These impurities are called doping agent or dopant.
- The increase in conduction is due to participation of impurity elements in between the valence band and conduction band.
- They make a bridge through which electrons can jump easily from the valence band to the conduction band.
- Doping is process of developing positive or negative charge through oxidation or reduction of the semiconductor.
- Doping are mainly two types:

$\left\{ \begin{array}{l} \text{Insulators: } \sigma < 10^{-8} \text{ S/cm} \\ \text{Semiconductors: } 10^{-7} < \sigma < 10^{-1} \text{ S/cm} \\ \text{Conductors: } \sigma > 10^2 \text{ S/cm} \end{array} \right.$

A) Poly(N-vinylcarbazole)



Photoconducting, conduct electricity a small degree under the influence of **light**

Used in the electrophotography (**photocopying**) industry.

C) Structural features for conduction

1. **Delocalization**: an **extended conjugated system** is usually necessary for backbone conductivity

2. Doping

Dopants:

- { Electron **acceptors**: e.g. AsF_5 or halogen
- { Electron **donors**: e.g. alkali metals

Conductivity varies with **dopant concentration**

3. Morphology:

Conduction is influenced by configurational and conformational factors, as well as **crystallinity**.

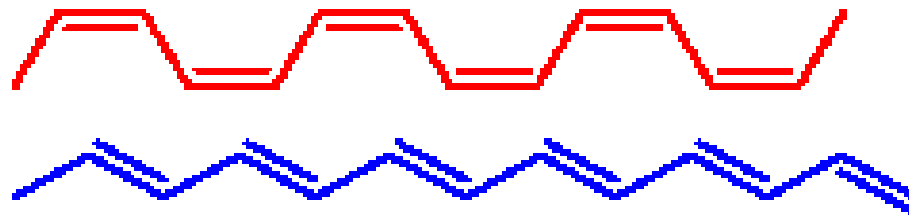
Conductivity of polyacetylene film in the direction of **molecular alignment** is increased significantly by stretching

Poly(sulfur nitride): $\sigma = 100 \text{ S/cm}$
 Superconductivity at $T < 0.3 \text{ K}$

Polyacetylene

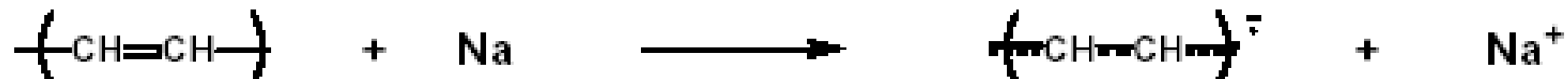
Cis $\sigma = 1.7 \times 10^{-9} \text{ S/cm}$

Trans $\sigma = 4.4 \times 10^{-5} \text{ S/cm}$



Doping apparently converts the **red metallic-colored cis** isomer to **blue metallic trans**.

Dopant forms a **charge transfer complex** with the polymer that give rise to **highly delocalized cation radicals** or **anion radicals** depending on whether the dopant is **electron accepting** or **donating**, respectively



The **delocalized regions**, called **solitons**, extend about **15 bond lengths**, at which point the **energy gain** arising from **lengthening of double bonds and shortening of single bonds** apparently outweighs **stabilization** arising from **delocalization**.



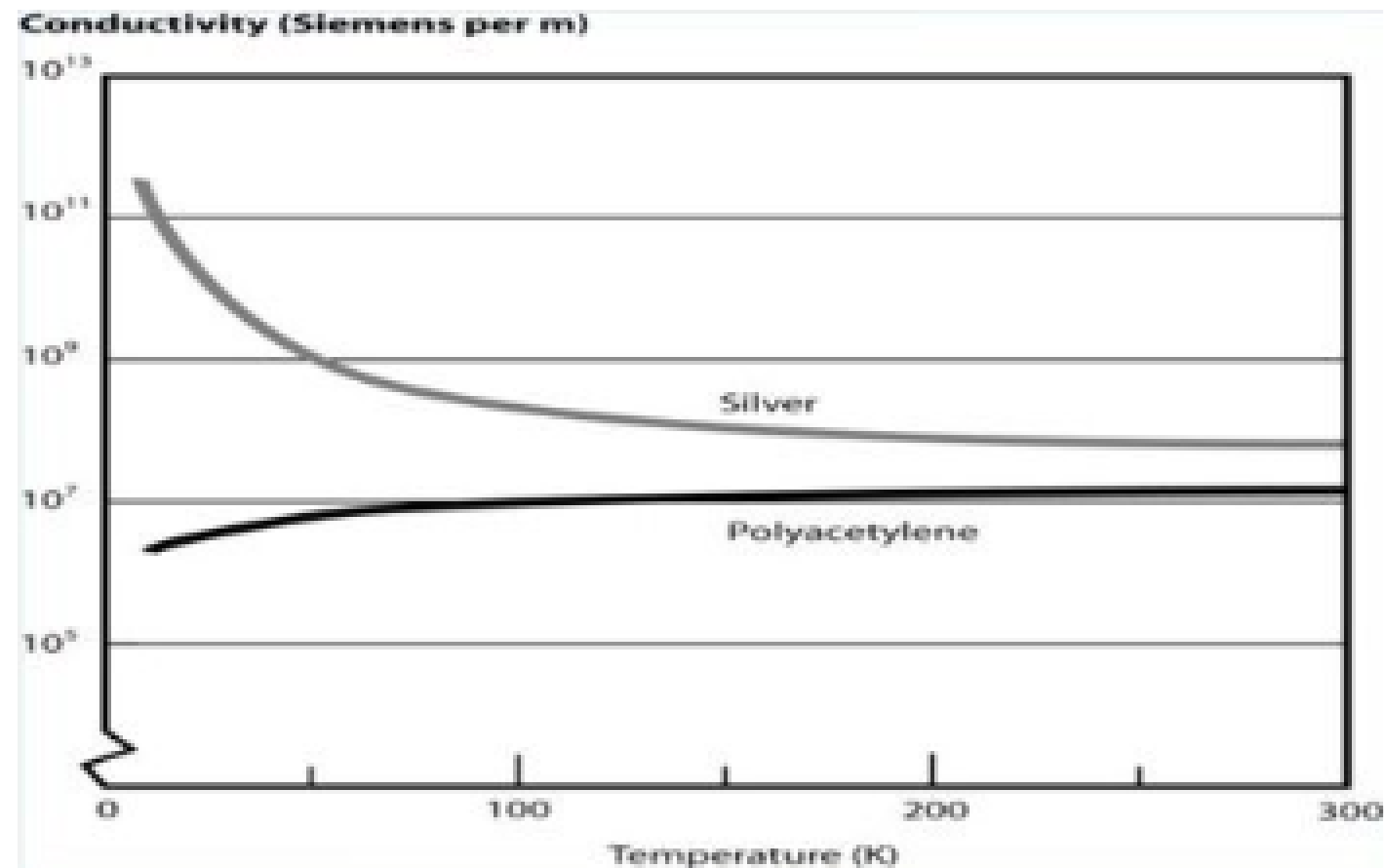
Soliton

Soliton: neutral (radical), positive (carbocation), or negative(carboanion).

p-type doping through oxidation of materials: In this type of doping some electrons from the conjugated π bonds are removed through oxidation creating a positive hole called **polaron** inside the polymer. The positive hole or polaron can move throughout the polymeric chain and make it conducting polymer.

n-type doping through reduction of materials: In this type of doping some electrons are introduced to the conjugated π bonds through reduction creating a negative hole or charge inside the polymer. The negative hole or charge can move throughout the polymeric chain and make it conducting polymer.

Effect of Temperature: Conductivity of a polymer increases with increase in temperature. Sometimes, the conductivity becomes constant at a particular temperature. The polymer formed at high temperature has higher conductivity and is stronger than that formed at lower temperature. Ex. Silver conductivity increase with falling temperature and polyacetylene conductivity increases with increasing temperature.

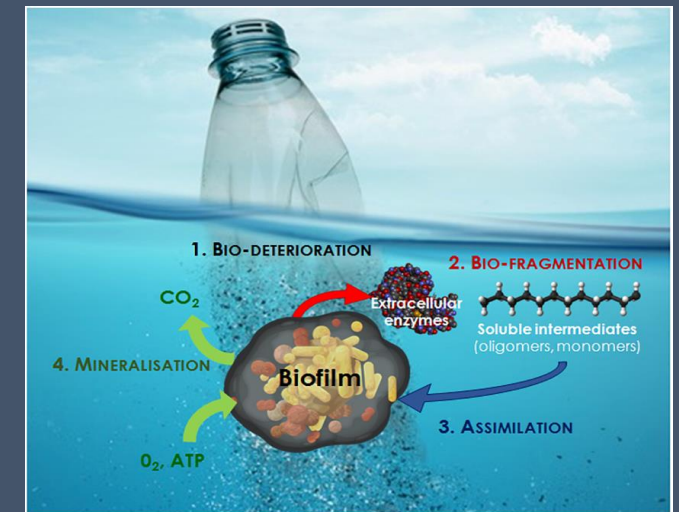
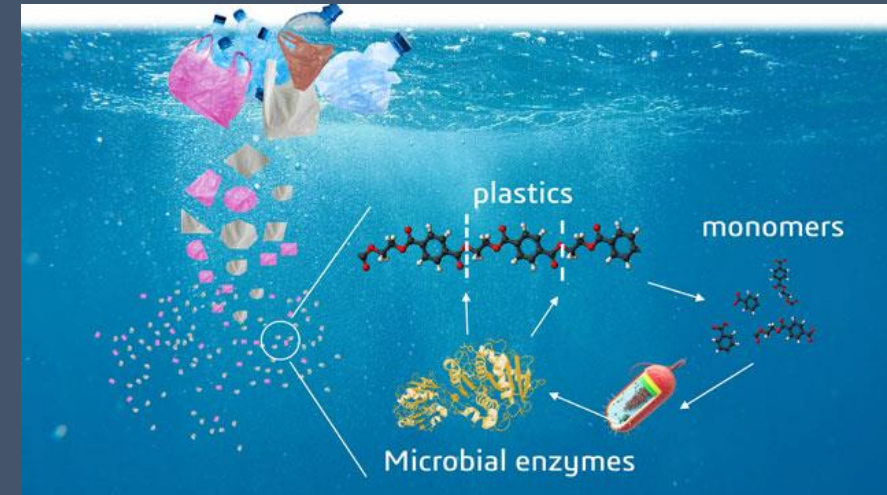


Content

Biodegradable Polymers: Definition and importance, examples

Factors controlling plastic biodegradation

Biodegradation steps



Biodegradable Polymers

Biodegradable polymers: They are a class of polymers that can be broken down into simpler substances by natural biological processes, such as those carried out by microorganisms. This breakdown typically results in harmless byproducts like carbon dioxide, water, and biomass. The chemical structure of these polymers is key to their biodegradability, with ester and amide linkages being particularly susceptible to hydrolysis and enzymatic degradation.

Importance: Biodegradable polymers offer a sustainable alternative to traditional plastics by breaking down into environmentally benign substances. Understanding the chemistry of these polymers, including their molecular structure and the factors that influence their degradation, is essential for developing and utilizing them effectively.

Concept: Plastic biodegradation is the microbial conversion of all its organic constituents to carbon dioxide, new microbial biomass and mineral salts under [oxic](#) conditions or to carbon dioxide, methane, new microbial biomass and mineral salts, under anoxic conditions.

Examples of Biodegradable Polymers

Bio-based polymers are three sub-categories:

- (i) Naturally occurring biopolymers (e.g. cellulose, starch, PHA and lignin),
- (ii) Bio-derived polymers (e.g. cellulose acetate),
- (iii) Synthetic polymers built from renewable (non-fossil) feedstock (e.g. polybutylene succinate (PBS)).

Aliphatic Polyesters: Examples include polylactic acid (PLA), polyglycolic acid (PGA), and polycaprolactone (PCL). These are often synthesized from renewable resources and are used in various applications, including packaging and medical implants.

Polyamides and Poly(ester-amide)s: Nylon 2-Nylon 6, a copolymer of glycine and aminocaproic acid, is an example of a biodegradable polyamide.

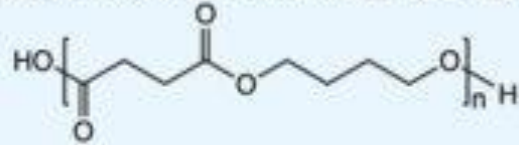
Polyhydroxyalkanoates (PHAs): A family of naturally occurring polyesters produced by microorganisms. Poly-3-hydroxybutyrate (PHB) is a common example.

Modified Natural Polymers: Cellulose, starch, chitin, and chitosan are examples of natural polymers that can be modified to enhance their biodegradability.

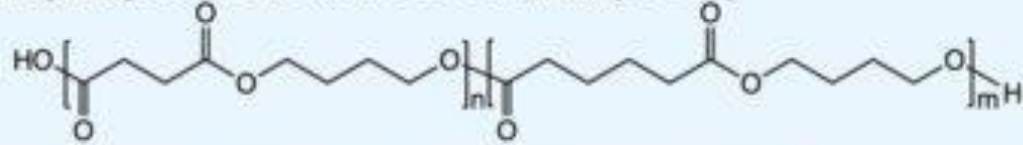
Polyurethanes: While some polyurethanes are not inherently biodegradable, specific formulations can be designed to be biodegradable.

a. Commonly used biodegradable polymers

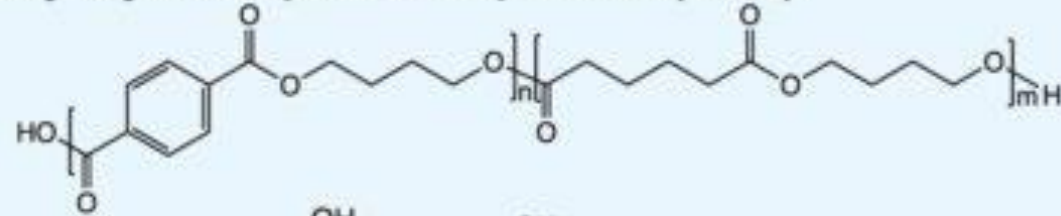
polybutylene succinate (PBS)



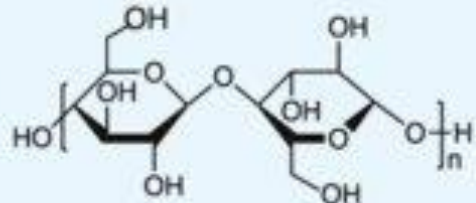
polybutylene succinate-co-adipate (PBSA)



polybutylene adipate-co-terephthalate (PBAT)



cellulose

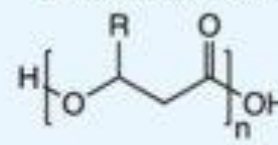


cellulose acetate

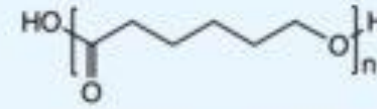
starch

thermoplastic starch

polyhydroxy-alkanoates (PHA)



polycaprolactone (PCL)



polylactic acid (PLA)

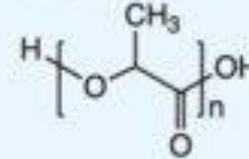


Figure a: Overview of commonly used biodegradable polymers with some chemical structures. Note that there are various variants of PBSA and PBAT; only a generic structure is shown.

b. schematic of polyester hydrolysis

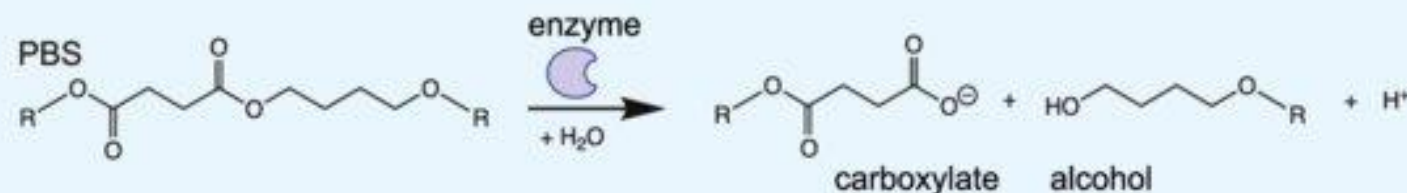


Figure b: Schematic of enzymatic hydrolysis of the aliphatic biodegradable polyester PBS under formation of the respective carboxylic acid and the alcohol.

Factors controlling plastic biodegradation

Microorganism-dependent

Polymer-dependent

Environment-dependent

Microbial Activity: The presence and type of microorganisms in the environment play a crucial role in the biodegradation process.

Environmental Conditions: Temperature, humidity, pH, and oxygen availability can affect the rate and extent of biodegradation.

Polymer Structure: The chemical structure of the polymer, including the type of linkages and the presence of branching or crystallinity, influences its susceptibility to degradation.

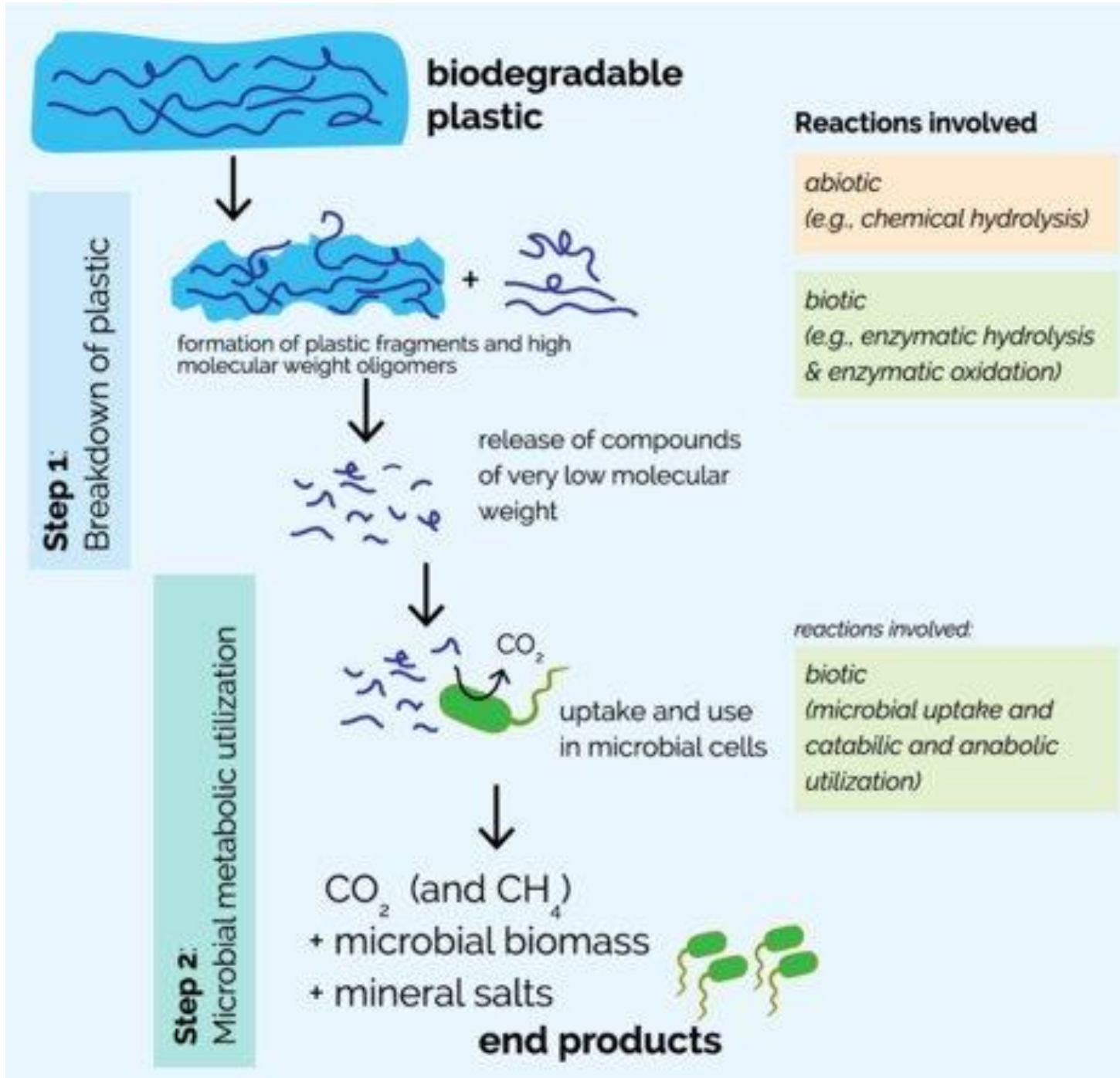
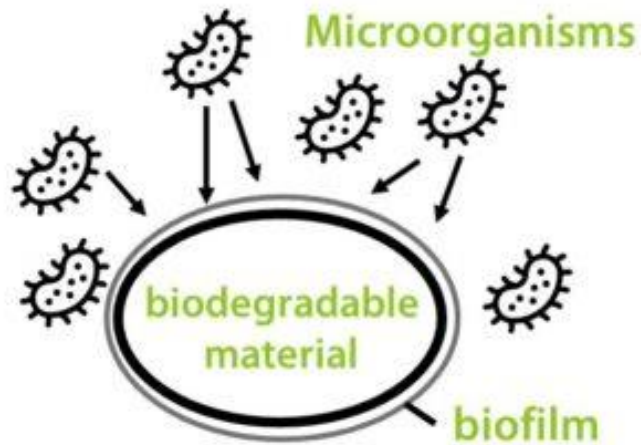


Figure : Overview of two key steps involved in plastic biodegradation.

Step 1 is the breakdown of the plastic into low molecular weight products that have sufficiently low molecular weights to be utilised by microorganisms.

Step 2 is the microbial utilisation of plastic degradation products formed in Step 1, under formation of CO₂ (or CO₂ and CH₄ under anoxic conditions) and microbial biomass.

How does biodegradable work?

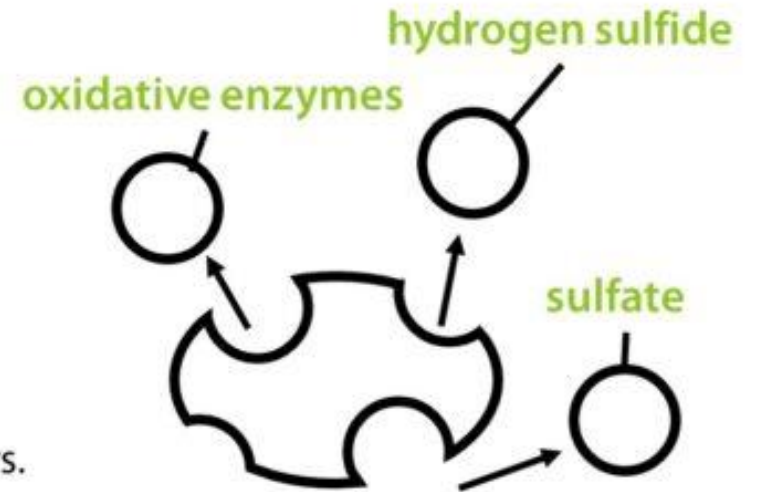
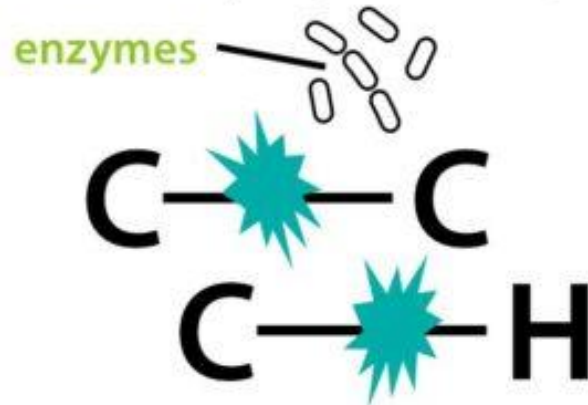


1 Forming biofilm on the surface

Microorganisms aggregate on the surface of biodegradable materials and produce sticky extracellular polymeric substances.

2 Depolymerization of enzymes

Extracellular polymeric substances include enzymes that can break down the carbon-carbon and carbon-hydrogen bonds of polymers.



3 Bioassimilation and mineralization

Become a source of carbon and energy for microorganisms or provide nutrition to the soil.